

Mannat Trivedi

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Executive Summary

Results-driven Software Engineer with a strong foundation in core computer science, system design, and full-stack development. Demonstrated capability in architecting high-performance, production-ready systems, optimizing latency, and integrating AI capabilities. Proven ability to deliver scalable solutions in both enterprise (Bosch) and independent environments.

Education

B.Tech in Computer Science and Engineering Maulana Azad National Institute of Technology (MANIT), Bhopal	Expected 2027 CGPA: 8.64
12th Grade (CBSE Board) RD Public School, Betul	2023 Score: 91.4%

Technical Skills

Core CS Fundamentals:	Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Memory Management, System Design
Languages:	C++, C, JavaScript, HTML, CSS, Tailwind CSS
Backend & Web:	Node.js, Express.js, React.js, Next.js, REST APIs
Databases:	MySQL, MongoDB, IndexedDB
Tools & Platforms:	Git, GitHub, Postman, MongoDB Compass, Ollama, Vercel, Railway, Render, Azure, SAP BTP Cloud Foundry

Experience

Software Engineering Intern Bosch Global Software Technologies (BGSW)	May 2026 – July 2026
- Developed and deployed an internal Employee Skill Assessment Platform for the TAF Department using React.js and Express.js, reducing assessment submission time by 40–50% through bulk uploads and dynamic form generation. - Implemented secure Microsoft Entra ID (Azure MSAL) authentication with Microsoft Graph API and SharePoint integration, improving enterprise data security, consistency, and centralized data management. - Engineered scalable backend services for reliable concurrent request handling and successfully deployed the application on SAP BTP Cloud Foundry.	

Selected Projects & System Architecture

The Binary Critic — Taste-based Media Discovery Platform	Next.js, Node.js, MongoDB, IndexedDB, NextAuth
Algorithmic Search: Engineered a Two-Stage Recommendation Engine (Retrieval → Multi-Signal Ranker → MMR), cutting feed latency by 67–80% (3.3s to 0.52s) by pairing a Metadata similarity algorithm with batched \$in aggregations that reduced DB round-trips from >50 to 2. Caching Hierarchy: Architected a 3-tier caching hierarchy (IndexedDB → Next.js ISR → MongoDB), achieving 580,000+ ops/sec client throughput (0.0017ms/op) and zero memory leaks via a Hybrid LRU + SWR eviction policy capped at 50 rails with indexed cursor-based pruning and 7-day TTL sweeps. Frontend Optimization: Slashed LCP by 86% (6.6s to 0.9s), TBT by 67% (120ms to 40ms), network payload by 95% (21MB to <1MB), and reducing CLS to 0.01 via layout-stable skeleton streaming, responsive Next/Image WebP delivery, and dynamic Open Graph structured metadata.	

QueryCraft — Conversational Data Analyst

Next.js, Node.js, MySQL, llama.cpp, Ollama, Groq, MiniLM

- **AI Pipeline:** Achieved 82.98% accuracy on the BIRD-SQL benchmark (Mini-Dev) Subset using a multi-step (Collect–Draft–Critic) reasoning pipeline for structured NL-to-SQL generation.
- **RAG Architecture:** Reduced schema hallucinations by ~90% by introducing local semantic schema retrieval with MiniLM embeddings and a lightweight Vectra-backed vector index.
- **Security:** Built an on-prem, LLM-agnostic, privacy-first system supporting local and cloud inference, with read-only enforcement and pre-execution validation for safe SQL execution.

InvenTrack — Inventory Management System

Node.js, Express, MySQL, JWT, bcrypt

- Engineered RBAC with JWT authentication and bcrypt password hashing, enabling permission-based access control rules.
- Built product management with image uploads, low-stock alerts, and CSV export, achieving a Lighthouse score of 99 with 0ms TBT.

TidyTrack: Built and deployed a campus issue tracking Android app for MANIT (tt.manit.ac.in).

MEENA: AI-powered student assistant built for Smart India Hackathon 2025 to reduce admin overhead (meenachatbot.vercel.app).

FraudRakshak: Cybersecurity dashboard/analyzer designed to detect fraudulent Android APKs (fraudrakshak.vercel.app).

Achievements & Personal Explorations

- **3rd Place:** CodeRush Web Development Contest (Developed a robust quiz platform under tight time constraints).
- **DSA Problem Solving:** Consistent practice across LeetCode mastering core data structures and advanced optimization patterns.
- **LIMO-DB & Resume2Web:** Developed a lightweight embedded C++ database engine and an instant static markdown-to-web resume engine.